dl.py - C:\Users\pings\Desktop\dl.py (3.14.2)

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```
#Exp No:15 Experiment to understand the data preprocessing in Data science
#Description: Understand the importance of Data preprocessing in data science

import numpy as np
import pandas as pd
pd.set_option('display.width', 200)
pd.set_option('display.max_columns', None)
df=pd.read_csv("pre-process_datasample.csv")
print(df)
df.info()
print(df.Country.mode())
print(df.Country.mode()[0])
print(type(df.Country.mode()))
df['Country']=df.Country.fillna(df.Country.mode()[0])
df['Age']=df.Age.fillna(df.Age.median())
df['Salary']=df.Salary.fillna(round(df.Salary.mean()))
print(df)
print(pd.get_dummies(df.Country))
updated_dataset=pd.concat([pd.get_dummies(df.Country),df.iloc[:,[1,2,3]]],axis=1)
print(updated_dataset)
df.info()
updated_dataset['Purchased']=updated_dataset.Purchased.replace(['No','Yes'],[0,1])
print(updated_dataset)
```

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IDLE Shell 3.14.2

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```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:\Users\pings\Desktop\dl.py ==========
   Country   Age   Salary Purchased
0   France  44.0  72000.0        No
1    Spain  27.0  48000.0       Yes
2  Germany  30.0  54000.0        No
3    Spain  38.0  61000.0        No
4  Germany  40.0      NaN       Yes
5   France  35.0  58000.0       Yes
6    Spain   NaN  52000.0        No
7   France  48.0  79000.0       Yes
8      NaN  50.0  83000.0        No
9   France  37.0  67000.0       Yes
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    9 non-null      str
 1   Age        9 non-null      float64
 2   Salary     9 non-null      float64
 3   Purchased  10 non-null     str
dtypes: float64(2), str(2)
memory usage: 532.0 bytes
0    France
Name: Country, dtype: str
France
<class 'pandas.Series'>
   Country   Age   Salary Purchased
0   France  44.0  72000.0        No
1    Spain  27.0  48000.0       Yes
2  Germany  30.0  54000.0        No
3    Spain  38.0  61000.0        No
4  Germany  40.0  63778.0       Yes
5   France  35.0  58000.0       Yes
6    Spain  38.0  52000.0        No
7   France  48.0  79000.0       Yes
8   France  50.0  83000.0        No
9   France  37.0  67000.0       Yes
   France  Germany  Spain
0    True    False  False
1   False    False   True
2   False     True  False
```

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```
4  Germany  40.0  63778.0       Yes
5   France  35.0  58000.0       Yes
6    Spain  38.0  52000.0        No
7   France  48.0  79000.0       Yes
8   France  50.0  83000.0        No
9   France  37.0  67000.0       Yes
   France  Germany  Spain
0    True    False  False
1   False    False   True
2   False     True  False
3   False    False   True
4   False     True  False
5    True    False  False
6   False    False   True
7    True    False  False
8    True    False  False
9    True    False  False
   France  Germany  Spain   Age   Salary Purchased
0    True    False  False  44.0  72000.0        No
1   False    False   True  27.0  48000.0       Yes
2   False     True  False  30.0  54000.0        No
3   False    False   True  38.0  61000.0        No
4   False     True  False  40.0  63778.0       Yes
5    True    False  False  35.0  58000.0       Yes
6   False    False   True  38.0  52000.0        No
7    True    False  False  48.0  79000.0       Yes
8    True    False  False  50.0  83000.0        No
9    True    False  False  37.0  67000.0       Yes
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 1   Age        10 non-null     float64
 2   Salary     10 non-null     float64
 3   Purchased  10 non-null     str
dtypes: float64(2), str(2)
memory usage: 536.0 bytes
   France  Germany  Spain   Age   Salary Purchased
0    True    False  False  44.0  72000.0         0
1   False    False   True  27.0  48000.0         1
2   False     True  False  30.0  54000.0         0
3   False    False   True  38.0  61000.0         0
4   False     True  False  40.0  63778.0         1
```

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```
   France  Germany  Spain
0    True    False  False
1   False    False   True
2   False     True  False
3   False    False   True
4   False     True  False
5    True    False  False
6   False    False   True
7    True    False  False
8    True    False  False
9    True    False  False
   France  Germany  Spain   Age   Salary Purchased
0    True    False  False  44.0  72000.0        No
1   False    False   True  27.0  48000.0       Yes
2   False     True  False  30.0  54000.0        No
3   False    False   True  38.0  61000.0        No
4   False     True  False  40.0  63778.0       Yes
5    True    False  False  35.0  58000.0       Yes
6   False    False   True  38.0  52000.0        No
7    True    False  False  48.0  79000.0       Yes
8    True    False  False  50.0  83000.0        No
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dtypes: float64(2), str(2)
memory usage: 536.0 bytes
   France  Germany  Spain   Age   Salary Purchased
0    True    False  False  44.0  72000.0         0
1   False    False   True  27.0  48000.0         1
2   False     True  False  30.0  54000.0         0
3   False    False   True  38.0  61000.0         0
4   False     True  False  40.0  63778.0         1
5    True    False  False  35.0  58000.0         1
6   False    False   True  38.0  52000.0         0
7    True    False  False  48.0  79000.0         1
8    True    False  False  50.0  83000.0         0
9    True    False  False  37.0  67000.0         1
>>>
```

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